Heart Disease Identification Method Using ML

A Project Report

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CERTIFICATE

This is to certify that the report entitled **Heart Disease Identification Method Using ML** submitted by **AISWARAYA A** (PKD19CS003),**KIRAN K** (PKD19CS028), **S SAYUJ** (PKD19CS047),**VISHNU C** (PKD19CS061) to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Computer Science AND Engineering is a bonafide record of the seminar work carried out by them under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

We AISWARAYA A ,KIRAN K ,S SAYUJ ,VISHNU C , hereby declare that the seminar report **Heart Disease Identification Method Using ML**, submitted for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of Joby N J

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

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Abstract

Heart disease is one of the complex diseases and globally many people suffered from this disease. On time and efficient identification of heart disease plays a key role in healthcare, particularly in the field of cardiology. In this article, we proposed an efficient and accurate system to diagnosis heart disease and the system is based on machine learning techniques. The system is developed based on classification algorithms includes Support vector machine, Logistic regression, Artificial neural network, K-nearest neighbor, Naive bays, and Decision tree while standard features selection algorithms have been used such as Relief, Minimal redundancy maximal relevance, Least absolute shrinkage selection operator and Local learning for removing irrelevant and redundant features.The features selection algorithms are used for features selection to increase the classification accuracy and reduce the execution time of classification system.The suggested diagnosis system (FCMIM-SVM) achieved good accuracy as compared to previously proposed methods. Additionally, the proposed system can easily be implemented in healthcare for the identification of heart disease.

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Abbreviation

Abbreviation	Description
ANN	Artificial neural network
SVM	Support vector machine
CVD	Cardiovascular diseases
MFCC	Mel Frequency Cepstrum Coefficient
CNN	Convolutional Neural Network
PCG	Phonocardiogram
TP	True positive
TN	True Negative
FP	False Positive
FN	False Negative
DFD	Dataflow Diagram
UML	Unified Modeling Language
HD	Heart Diseases
HRFLM	Hybrid Random Forest with Linear Mode
PCA	Principal Component analysis height

Chapter 1

Introduction

Heart disease (HD) is the critical health issue and numerous people have been suffered by this disease around the world. The HD occurs with common symptoms of breath shortness, physical body weakness and, feet are swollen . Researchers try to come across an efficient technique for the detection of heart disease, as the current diagnosis techniques of heart disease are not much effective in early time identification due to several reasons, such as accuracy and execution time. The diagnosis and treatment of heart disease is extremely difficult when modern technology and medical experts are not available. The effective diagnosis and proper treatment can save the lives of many people. Diagnosis of HD is traditionally done by the analysis of the medical history of the patient, physical examination report and analysis of concerned symptoms by a physician. But the results obtained from this diagnosis method are not accurate in identifying the patient of HD. Moreover, it is expensive and computationally difficult to analyze . Thus, we have to develop a noninvasive diagnosis system based on classifiers of machine learning (ML) to resolve these issues. A machine learning based diagnosis method were proposed for the identification of HD in this research work. Machine learning predictive models include ANN, LR, K-NN, SVM,DT, and NB are used for the identification of HD.

Chapter 2

Literature review

2.1 Predicting the Presence of Heart Diseases using Comparative Data Mining and Machine Learning Algorithms

2.1.1 Introduction

Heart diseases such as heart failure,and myocardial infarction have been ranked as the highest cause of death in the United States. According to the center for disease control and prevention, about 610,000 people die every year from heart disease. Health professionals have put measures in place for early detection of heart diseases, as this is key in preventing and curbing them. However, early-stage adopted strategies in identifying heart diseases have not been successful, due to the associated complexities. Unlike traditional statistical methods, data mining techniques can detect and extract hidden inconspicuous patterns, and relationships in large datasets. Support vector machine, Na¨ıve Bayesian, artificial neural networks, logistics regression, etc. models have been developed and used in healthcare research. They have shown immense potential in the accurate prediction of heart diseases based on clinical data of patients.

2.1.2 Method

This research uses data exploration and mining techniques to extract hidden patterns using Python. In this research different machine learning algorithms are used like -

logistic linear regression, decision tree classifier, and Gaussian Naive Bayes model which are developed to predict the patient's presentation of heart disease. It seeks better performance in heart disease prediction in order to reduce the number of tests required for heart disease diagnosis. A k-fold cross-validation approach is used in evaluating the model resulting in receiver operating characteristic (ROC) curves.

The model uses a 10-fold cross-validation method resampling method was adopted in the training and testing of the model. The dataset was split into 10 folds. The first iteration uses the data in the 1st fold to test the model while the remaining 9 folds are used to train the model. The second iteration uses the 2nd fold as the test set and the remaining 9 as the training set.

The system has 5 steps - data preprocessing, data transformation, feature reduction, model training, and evaluation of results.

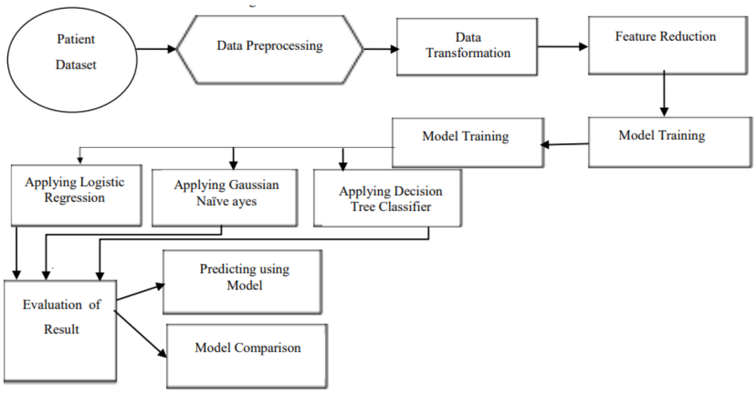

Figure 2.1: Block diagram of the ML model

The three classification models were analyzed by evaluating the accuracy, recall, f1, and precision scores. The performance of each classification model on test data was visualized using the confusion matrix

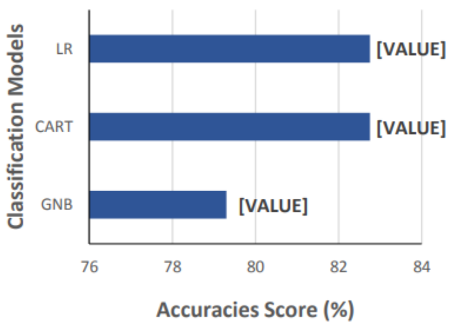

Figure 2.2: Accuracies scores of each algorithm

2.1.3 Avantages

- Identify the most effective solution for a particular problem
- Easy understanding of working of models

2.1.4 Disadvantage

- Time-consuming
- relies on a fixed set of data and models, and may not be able to identify the best solution if the true underlying data-generating process is different from the one used in the comparison.

2.2 Accurate classification of heart sound for disease diagnosis by a single time varying spectral feature: Preliminary Results

2.2.1 Introduction

Among the causes of global death, cardiovascular diseases (CVDs) are ranked number one by the World Health Organization (WHO). They account for 31 percentage (17.9 million) of total worldwide deaths . Therefore, it is critical to diagnose these diseases at an early stage in order to avoid or minimize their undesirable consequences.

Early detection of CVDs is not a trivial task and it requires the application of generally invasive methods that can only be conducted by medical experts with appropriate training.While the accuracy of this approach regarding the diagnosis of CVDs is very high, its cost is also very high making its accessibility limited.It is very difficult and in most cases highly inaccurate to use the segmented heart sounds for classification for the purpose of disease diagnosis. In order to alleviate this difficulty, classification is usually not based on signal samples of segmented heart sound recordings, but based on a number of features extracted from these recordings.Therefore, precise feature extraction is key to accurate classification of heart sounds and plays a major role in the performance of the subsequent classification stage .

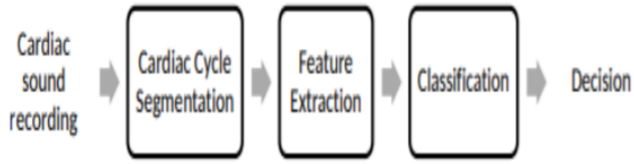

Figure 2.3: Main components of an automatic heart sound analysis system

2.2.2 Method

Figure2.4 shows the block diagram of an automatic heart sound analysis system. First, the heart sound recording is fed to a segmentation block. Here, the signal is segmented into individual cardiac cycles and these cycles are further segmented into main heart sound segments. Next, these segmented signal reco rdings are processed by a feature extraction block to produce one or more features that are expected to better represent the heart sound signal. Finally, these features are used to drive a classifier block, which generates a diagnostic decision at its output.

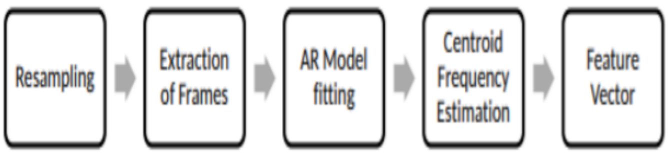

Figure 2.4: Steps involved in feature extraction.

2.2.3 Advantages

- Early detection of abnormal heart conditions helps reduce sudden cardiac deaths caused by cardiovascular diseases (CVDs).
- It enables the classifier to perform the classification by using only one feature.
- The SVM classifier also performed well.

2.2.4 Disadvantages

- Depending on the size of dataset and type of features in the classification experiment different results can be achieved with variant accuracie.

2.3 Effective Heart Disease Prediction Using Hybrid Machine Learning Techniques

2.3.1 Introduction

It is difficult to identify heart disease because of several contributory risk factors such as diabetes, high blood pressure, high cholesterol, abnormal pulse rate and many other factors. Various techniques in data mining and neural networks have been employed to find out the severity of heart disease among humans. The severity of the disease is classified based on various methods like K-Nearest Neighbor Algorithm (KNN), Decision Trees (DT), Genetic algorithm (GA), and Naive Bayes (NB).Neural networks are generally regarded as the best tool for prediction of diseases like heart disease and brain disease. The proposed method which we use has 13 attributes for heart disease prediction.Hybrid Random Forest with Linear Model (HRFLM) is introduced to improve the performance accuracy of heart disease prediction,HRFLM method uses all features without any restrictions of feature selection.Hybrid method has stronger capability to predict heart disease compared to existing .

2.3.2 Method

In HRFLM, we use a computational approach with the three association rules of mining namely, apriori, predictive and Tertius to find the factors of heart disease on the UCL Cleveland dataset.HRFLM makes use of ANN with back propagation along with 13 clinical features as the input.Data mining methods help in remedial situations in the medical field. The data mining methods are further used considering DT, NN, SVM, and KNN. Among several employed methods, the results from SVM prove to be useful in enhancing accuracy in the prediction of disease.The performance efficacy of this method can be estimated from the accuracy in the outcome results based on ECG data. ANN training is used for the accurate diagnosis of disease and the prediction of possible abnormalities in the patient.The Probabilistic Principal Component Analysis (PPCA) method is proposed for evaluation, based on three data sets of Cleveland, Switzerland, and Hungarian in UCI respectively. The method extracts the vectors with high covariance and vector projection used for minimizing the feature dimension. The feature selection with minimizing dimension is provided to a radial basis function, which supports kernel-based SVM.The feature selection plays a prominent role in the prediction of heart disease. ANN with back propagation is proposed for better prediction of the disease. The results obtained from the application of ANN are highly accurate and very precise.The genetic algorithm with fuzzy NN known as Recurrent Fuzzy Neural Network (RFNN) is introduced for the diagnosis of heart disease.

2.3.3 Advantages

- Improved accuracy in the prediction of cardiovascular disease.
- HRFLM method uses all features without any restrictions of feature selection.

2.3.4 Disadvantages

- Large number of trees can make the algorithm too slow and ineffective for real-time predictions.
- Random forest are fast to train, but quite slow to create predictions once they are trained.

2.4 Cardiovascular Disease Prediction Using Machine Learning models

2.4.1 Introduction

According to WHO, every year, twelve million losses of life happen globally due to Heart diseases.. It is one of the main reasons for death in various countries. It has considered the primary reason for death in adults. Coronary heart disease and Cardiovascular disease are classes of heart diseases. The term cardiovascular disease includes several contingencies that affect the heart, blood vessels, and circulating and pumping blood throughout the body. Cardiovascular diseases result in several illnesses, disabilities, and death. The diagnosis of diseases is an essential and complex responsibility in medicine.

Using machine learning algorithms, we can recognize the disease at an initial stage and help to cure disease with a conventional diagnosis. To develop the model, we have used different machine learning algorithms to resolve the problem; we also tried to find the significant factor while predicting cardiovascular disease. Various issues, such as absent value, outlier, and recognizing important dimensions, are well handled by decision trees.

2.4.2 Method

According to Figure2.5 the system is implemented using five major steps in prediction cardiovascular disease.

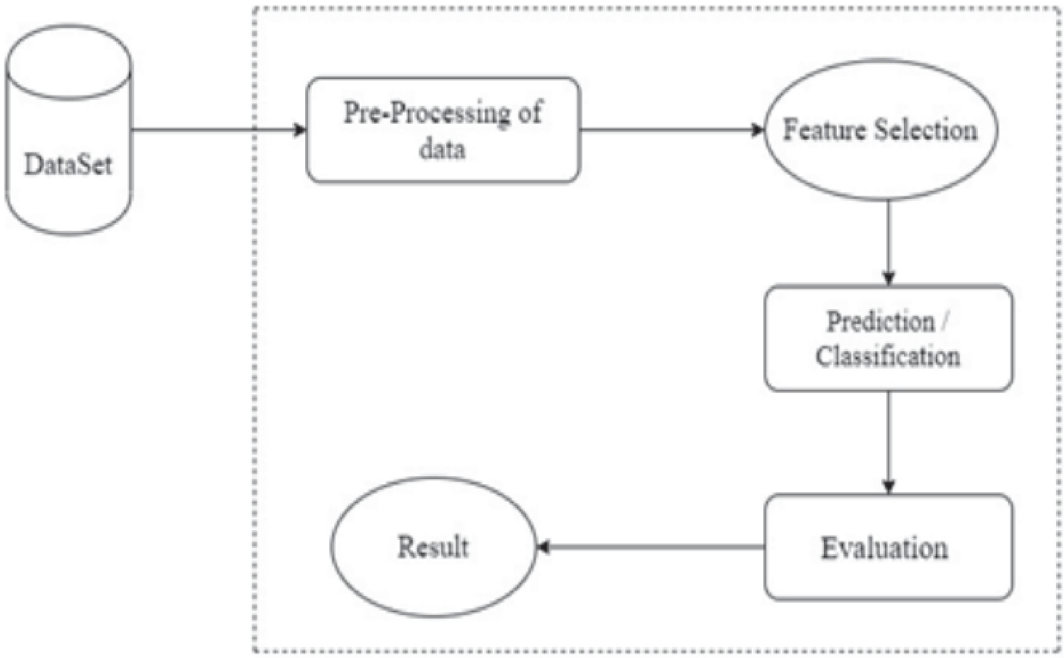

Figure 2.5: Experiment Workflow with dataset

The major steps are:

- Data Description
- Data Pre-processing
- Feature Selection and Adding a new feature
- Experimental setup for evaluation
- Classification

Here classification is used in the prediction of heart disease. Various machine learning models like logistic regression,k-nearest neighbour,naive bayes,neural network,decision tree etc are used.The classification models were developed using 13 features. The train and test accuracies were calculated for each model. The evaluation of the model was based on seven different classifiers. After Classifying both models, A and B, BMI is a significant factor to increase the accuracy of the model.

2.4.3 Advantages

- The decision classifier approach proved to be very efficacious to predict the diseased using features like age, BMI, cholesterol etc.

2.4.4 Disadvantages

- For some algorithms, the testing accuracy is slightly greater than training accuracy. Generally, the testing accuracy should be less than that of the training accuracy.

2.5 Heart Disease Prediction System Using Model Of Machine Learning and Sequential Backward Selection Algorithm for Features Selection

2.5.1 Introduction

Detection and treatment of HD is difficult where modern diagnosis technology and medical experts are not available in developing countries .The effective diagnosis and appropriate treatment can save the lives of many people. sequential backward selection (SBS) feature algorithm to choose related features for better classification accuracy of K-NN ML. Cleveland heart disease dataset, 2016 is used.The K-NN performances evaluated with selected features. Finally, we suggest that sequential backward selection features algorithm to choose the appropriate features that should be used for better classification accuracy using K-Nearest Neighbor classifier.

2.5.2 Method

Dataset

Pre-processing Method

Preprocessing methods includes deletion values missing features, Standard Scalar, Min-Max Scalar. In standard Scalar method feature has mean and variance values

0 and 1. All features coefficient have same values. Min-Max Scalar feature values range between 0 and 1. Feature having missing values are deleted

Sequential backward selection (SBS)

SBS is a classical feature selection algorithm, which deduced the feature space into subspace feature with minimum latency in classifier performance and reduces the model execution time.To calculate that which feature should be eliminating from feature space at each step required to define a function of criterion J to minimize. The criterion is computed by criterion that is simply be the difference in performance of the classifier before and after the elimination of a specific feature .The feature that eliminated at each step can be defined as the feature that maximizes the criterion

Machine learning classifier

In this study K-NN ML algorithm used for detection of HD, K-NN predicts label of class a new input. It use similar value of new input its inputs samples of set in the training and if new input sample same in the set of training then performance is not good. Suppose x and y be the observations of training and function of learning is h: x→ y, Such that given observation x, h(x) can compute y value. In our experiments different number of features from subspace features will pass to K-NN for training and testing purpose. The output predictions of two class values of K-NN are HD and healthy subjects. For each number of subspace features k nearest neighbors were used. K is integer number values. 8 various k values were used for k = 1, 2, 3, 4, 5, 6, 7, 8 and mean value was calculated evaluate the performance with each number of selected features K-NN is good classifier because it no need extra parameters.

Validation method of classifier

To check classifier performance train /test validation method was used in the study. For each number of reduced features space 70On test validation score the Performance of K-NN was computed

Evaluation Metrics for K-NN

Performance metrics of evaluation used for K-NN performance measuring. Confusion metrics were used and each observation in set of testing is computed in exactly one box and it is 2x 2 matrixes because there are 2 repose classes Here are two classes' people with HD and health subjects in order to classify it; the following metrics can be calculated in this study.

TP (True Positive): It concluded that HD people are correctly detected which determined that subjects have HD patients.

TN (True Negative): It means accurate detection of healthy subjects and subjects have no HD.

FP (False Positive): It means the wrong detection of healthy subjects and the subjects have HD.

FN (False Negative): It means that HD is wrongly detected and the subjects are healthy.

Positive case is 1 for disease and negative case 0 for healthy.

2.5.3 Advantages

- High accuracy
- It will effectively identify the HD and healthy subjects.

2.5.4 Disadvantages

- Accuracy does not holds good for imbalanced data.
- Require high memory – need to store all of the training data.

2.6 Multiple Heart Diseases Prediction using Logistic Regression with Ensemble and Hyper Parameter tuning Techniques

2.6.1 Introduction

Heart disease can be predicted with more accuracy if preprocessing of data is done and then trained the ML model with that data. Various preprocessing techniques are used.Data is analyzed and checked for its distribution,balanced/imbalanced, skewness, kurtosis and appropriate methods are used to transform the data to normal distribution.

Feature engineering is done to get the important features for prediction and removed the less contributing features so that computation time of the model can be reduced and also data dimension can be reduced using PCA and Kernal PCA.Hyper parameter tuning,tunes the parameter's value of ML model to get high accuaracy. Using all these techniques, we have designed few ML models and predicted the heart disease.

2.6.2 Method

Implementation Environment

We have used Aanaconda4 (version 1.9.7) for implementation, python 3.7 environment. Data science has Anaconda as one of the standard platforms.

Dataset Analysis

We have taken Data set from UCI ML repository. Data consists of 303 patient's information which includes 14 features. The correlation between the features is shown through Heat map in Figure2.6. We can observe that features have positive,negative and zero correlation with other features.

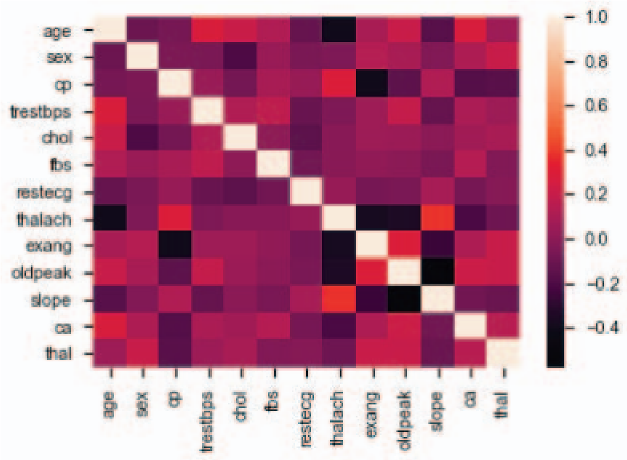

Figure 2.6: Heat map of the features

Pre-Processing

Transformation : To distribute the data normally various transformation techniques such as Standard scalar,Robust scalar and Power Transform are used. Among three transformations, power transform gives the better results so it is used in all the models .

Removal of outliers :Turkey fence approach is used to remove the outliers.

Feature Engineering

The steps involved in feature engineering are :

- Feature selection
- Extra Trees Classifier
- Dimension Reduction
- kernel PCA
- Feature Importance

Logistic Regression

Logistic regression is used as main ML model. Its a classification algorithm used to classify observations to a different set of classes. The sigmoid function, which is

S shaped curve, is used in this algorithm, which fits any practical value into range between 0 and 1.

2.6.3 Advantages

- Dimension reduction technique - Kernel PCA used to reduce the features as low as 5 features from 13 features to improve model effectiveness not only accuracy, but also computational performance.
- The model not only gives 100 percent accuracy, but it also gives a perfect score in terms of the AUC-ROC curve, Precision, recall, and other matrices of the model.

2.6.4 Disadvantages

- It should not be used if the number of observations is lesser than the number of features, otherwise, it may lead to overfitting.

2.7 Elimination Of Irrelevant Features And Heart Disease Recognition By Employing Machine Learning Algorithms Using Clinical Data

2.7.1 Introduction

Heart disease is a great medical issue and numerous people suffered from it around the world.The diagnosis and treatment of HD is difficult, particularly in the poor countries.According to the report of European Society of Cardiology that 26 million children universal were identified with HD. To the identifying HD through invasive based methods are founded on the examination of patient's health history and analysis of related signs by health specialists. These methods are not suitable for HD diagnosis.

In this study a new HD diagnosis method has been proposed. Feature selection algorithm Chi squared was used for related feature selection to improve the performance of the classifier from Cleveland data. Five machine learning models were used to predict HD. Model evaluation metrics have employed for model evaluation. The performance proposed method was compared to state of the art method.

2.7.2 Method

Data Set

The Cleveland heart disease dataset has been used in this model training and testing.

Feature Selection Based on Chi-squared Algorithm

The foundation is that if a feature is autonomous of the target variable it is uninformative for classifying clarifications. In feature selection, features are selected which are highly dependent on the response. When two features are independent, the observed count is close to the expected count, thus we will have smaller Chi-Square value. So high Chi-Square value indicates that the hypothesis of independence is incorrect.

Classification models

Logistic regression

It is incorporated mostly in classification related problems. It is used for binary machine learning problem of classification to predict the value of the predictive variable. Logistic regression is a data analysis technique that uses mathematics to find the relationships between two data factors. It then uses this relationship to predict the value of one of those factors based on the other. The prediction usually has a finite number of outcomes, like yes or no.

Decision tree

A decision tree is a type of supervised machine learning used to categorize or make predictions based on how a previous set of questions were answered. The model is a form of supervised learning, meaning that the model is trained and tested on a set of data that contains the desired categorization.The basic algorithm used in decision trees is known as the ID3 (by Quinlan) algorithm. The ID3 algorithm builds decision trees using a top-down, greedy approach.

Support Vector Machine (SVM)

Support Vector Machine(SVM) is a supervised machine learning algorithm used for both classification and regression. The objective of SVM algorithm is to find a hyperplane in an N-dimensional space that distinctly classifies the data points.Support vectors are data points that are closer to the hyperplane and influence the position and orientation of the hyperplane. Using these support vectors, we maximize the margin of the classifier. Deleting the support vectors will change the position of the hyperplane. These are the points that help us build our SVM.

2.7.3 Advantages

- Feature selection process has been performed by using Chi squared feature selection algorithm.Chi-square tests enable us to compare observed and expected frequencies objectively.
- High prediction accuracy
- Effective diagnosis od heart disease and can be easily incorporated in health care.

2.7.4 Disadvantages

- Difficulty of interpretation when there are large numbers of categories (20 or more) in the independent or dependent variables.
- In cross validation performance evaluation is subject to higher variance given the smaller size of the data.

2.8 Early Detection of Heart Valve Disease Employing Multiclass Classifier

2.8.1 Introduction

Cardiac auscultation is a technique of observing heart sound with a stethoscope.Once a cardiac auscultation signal is obtained, this may be classified through various software techniques but these techniques require highly precise well defined heartbeat signals for a useful feature extraction process. This paper presents a signal processing based heart sound analysis to distinguish between five classes of abnormal and normal heart sounds which can be useful for identification of cardiac diseases. The major finding and contribution of this communication is a distinguished analysis of heart sounds vibrations obtained from the healthy and unhealthy subjects for automatic screening of cardiovascular disorders.In order to understand and interpret the variations in the cardiac sound of disordered heart some characteristics such as Log filter bank Energies and frequency contents are studied and analyzed to automate the classification procedure.

2.8.2 Method

The stages associated with preparing and distinguishing the auscultation signals are the collection of heart sound, pre-processing steps such as noise removal, modeling of a classifier, testing the classifier using PCG. Process of modeling a classifier includes unique feature extraction in case of a normal or abnormal condition, preparing training data to machine learning algorithms and characterization of deceases. From heart sound we extracted six MFCC, LogFBank and Statistical features from the heart signal and these features are used to prepare a labeled database. Using database prepared at this level SVM based classifier is trained SVM (trained) model is deployed to classify a real-time Heart sound .

Recording of Phonocardiogram (PCG) signals

A digital stethoscope along with audio jack to USB typeinterface is used to record heart sounds. The recording ofheart sound signals is done using open source software “Audacity”

Preprocessing of recorded Phonocardiogram (PCG) signals

Recorded heart sound is corrupted and mixed by various causes of noise such as breathing noise, power line interference and motion artifacts etc.Removal of such components is done in preprocessing phase

Feature Extraction

Mean and median of MFCC for prediction of heart disease using SVM. heart sound is also considered to be a low-frequency audio signal, hence can be better classified with MFCC features .MFCCs are calculated via a method analogous to the cepstral notation of an audio signal, therefore, they may be preferred to provide a close approximation of the human hearing system.Formula used to calculate mel frequencies is

Mel (freq) = 2595*(log (1+freq/700).

Heart sound Classification using Multiclass Support Vector Machine (SVM)

A "support vector machine" (SVM) is a superior method for identification of pattern present in a set of data points. Support Vector Machine model in the classification of data points utilizes Lagrange Multipliers, in this approach overall pattern classification problem reduced to the fitting of a function with minimum error in the successful classification of the training data.The Support Vector Machine Model works equally well for both linearly separable problems and non linearly separable problems.

2.8.3 Advantages

- High SVM classification score is 97.50
- Efficient for classification of the multi-class heart sounds.
- Highly precised.

2.8.4 Disadvantages

- SVM classification does not execute very well when the data set has overlapping sounds.
- Training complexity of SVM is very high.

Chapter 3

PROPOSED SYSTEM

3.1 Proposed System

Three classification models namely; Linear Regression (LR), Decision Tree Classifier (CART) and Gaussian Naïve Bayes (GNB) were developed. The data was analyzed and implemented in python. Data preprocessing techniques such as, feature transformation,training, testing with the individual models, and finally comparison of the performance of the models were the steps followed by the system.The system compared the accuracies of the three algorithms (CART, LR and GNB) in predicting heart diseases in patients. The highest predicting accuracy score was obtained with the CART and GNB. They both had a predicting score of 82.75 percenttage,precision recall,f1 scores. However, the AUCROC value for the GNB was higher than the LR model. The CART model scored the highest predicting accuracy, among the three models. The least accuracy score obtained with the LR algorithm could be due to the relatively smaller sample size of the dataset.The greater AUCROC value (0.87) from GNB model makes it a better choice than LR (0.86). But overall CART model makes the final choice for the proposed system due to high accuracy comparitively.

3.2 Block Diagram

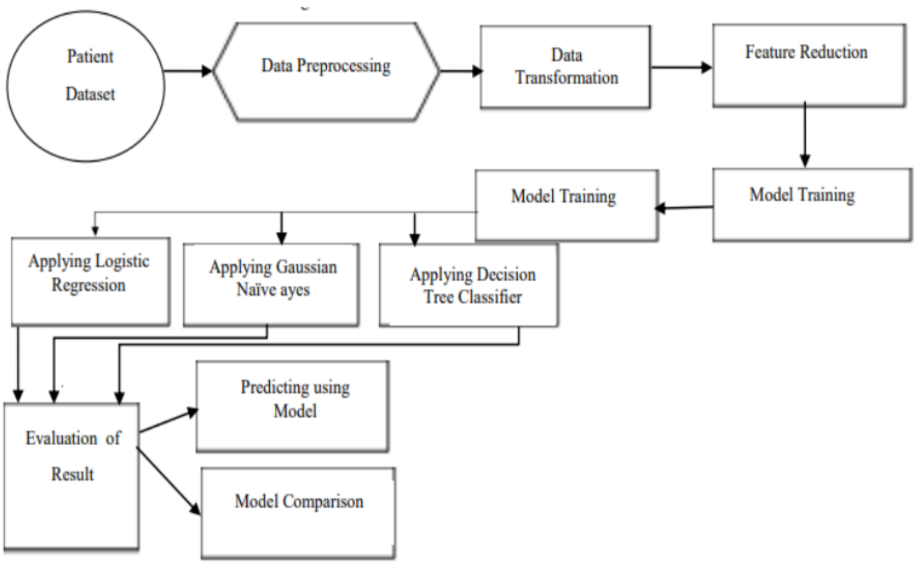

Figure 3.1: high-level block diagram summarizing the methodology adopted

Chapter 4

System Design

4.1 System Design

System design is the process of designing the elements of a system such as the architecture, modules, and components, and their interactions, to satisfy specified requirements. System design can be seen as the application of systems theory to create a model of a system. This model is then used to guide the development of the system, which can include hardware and software components. There are many different approaches to system design, and the specific approach used can depend on the specific system being designed, as well as the goals and constraints of the design process. Some common considerations in system design include performance, scalability, reliability, security, and maintainability.

4.1.1 Dataflow diagram

A data flow diagram (DFD) is a graphical representation of the flow of data through an information system, modeling its process aspects. A DFD is often used as a preliminary step to create an overview of the system.DFDs can also be used for the visualization of data processing (structured design). A DFD shows what kind of information will be input into and output from the system, where the data will come from and go to, and where the data will be stored. It does not show information about the timing of processes, or information about whether processes will operate in sequence or in parallel.

A DFD consists of four main elements:

- Processes: Represented by circles, these are the actions or transformations that take place on the data.
- Data stores: Represented by rectangles, these are the locations where data is stored either temporarily or permanently.
- Data flows: Represented by arrows, these show the flow of data from one element to another.
- External entities: Represented by ellipses, these are the sources or destinations of data that are external to the system being modeled.

Figure 4.1: DFD LEVEL 0

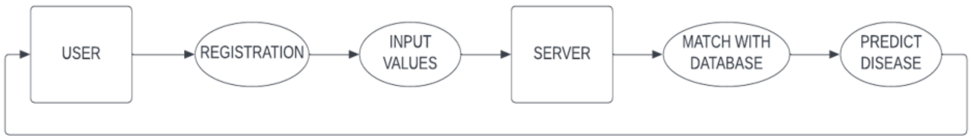

Figure 4.2: DFD LEVEL 1

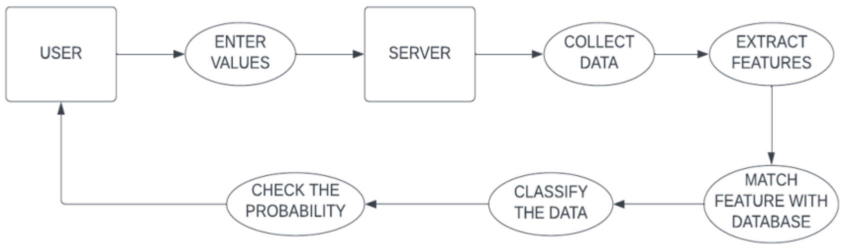

Figure 4.3: DFD LEVEL 2

4.1.2 UML Diagrams

Unified Modeling Language (UML) is a standardized general-purpose modeling language in the field of software engineering. The UML provides a way to visualize a system's architectural blueprints in a visual diagram, including elements such as:

- Classes and their attributes, operations, and relationships
- Objects and their relationships
- Use cases and scenarios
- State diagrams
- Activity diagrams
- Sequence diagrams

UML diagrams are used to model the design of a system, including its behavior, structure, and implementation. They can be used to communicate design decisions to developers and stakeholders, and to validate the design of a system.

4.1.3 use case diagram

A use case diagram is a type of UML diagram that shows the interactions between a system and its users (called actors) and the dependencies among those interactions (called use cases). Use case diagrams are used to represent the functionality of a system and to identify the requirements for that system. A use case diagram consists of the following elements:

- Actors: Represented by stick figures, actors are the external entities that interact with the system. They can be humans or other systems.
- Use cases: Represented by ovals, use cases are the interactions between actors and the system. They represent the functionality provided by the system to the actors.
- System boundary: Represented by a rectangle, the system boundary encloses the use cases and actors and defines the scope of the system being modeled.

- Associations: Represented by lines connecting actors and use cases, associations show the relationships between actors and use cases.

Use case diagrams are useful for identifying the requirements for a system and for communicating the functionality provided by the system to stakeholders. They can also be used to identify and model the relationships between actors and use cases, and to define the boundaries of the system being modeled.

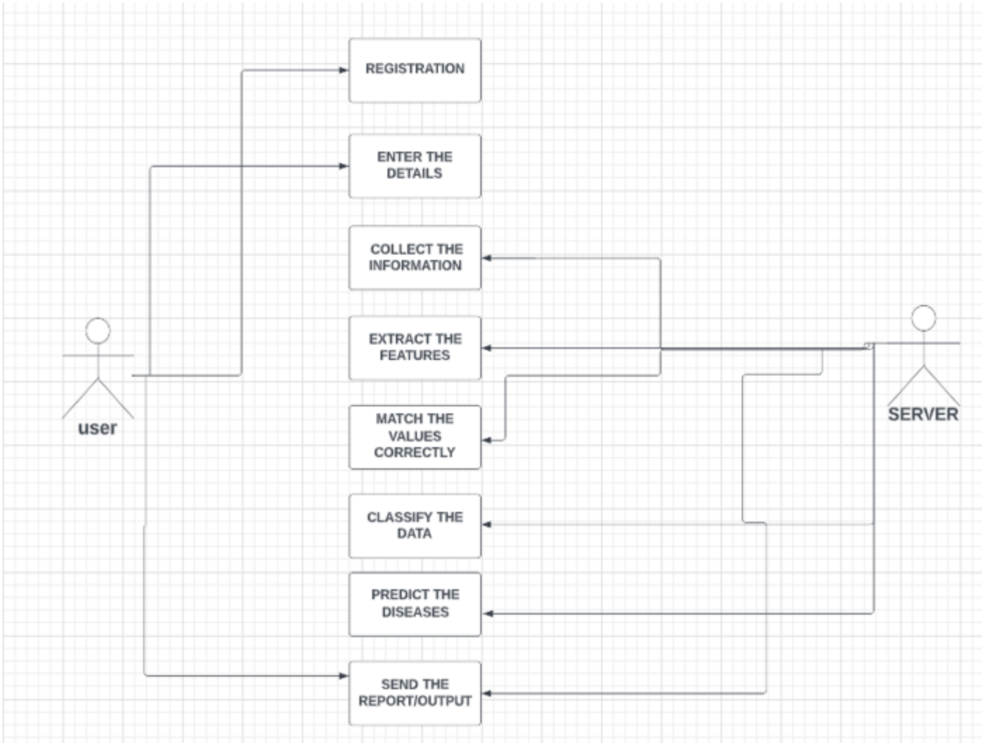

Figure 4.4: Use case diagram

4.1.4 Activity Diagram

An activity diagram is a type of UML diagram that shows the flow of activities within a system. It represents the actions that take place within the system and the order in which they occur. Activity diagrams are used to model the workflow behind the screens of a system, and to capture the logic of a business process.

An activity diagram consists of the following elements:

- Activity: Represented by a rounded rectangle, an activity represents a task or unit of work within the system.

- Initial node: Represented by a solid circle, an initial node indicates the start of a process.
- Final node: Represented by a solid circle with a bullseye, a final node indicates the end of a process.
- Decision node: Represented by a diamond, a decision node represents a branching point in the process, where the flow of control is directed based on the outcome of a decision.
- Fork and join: Represented by horizontal and vertical bars, a fork indicates a point in the process where the flow splits into multiple parallel paths, and a join indicates a point where the multiple parallel paths converge.
- Transitions: Represented by arrows, transitions show the flow of control from one activity to another.

Activity diagrams are useful for modeling the workflow behind the screens of a system and for understanding the logic of a business process. They can be used to model the flow of control in a process, to identify potential problems in the process, and to optimize the process for efficiency and effectiveness.

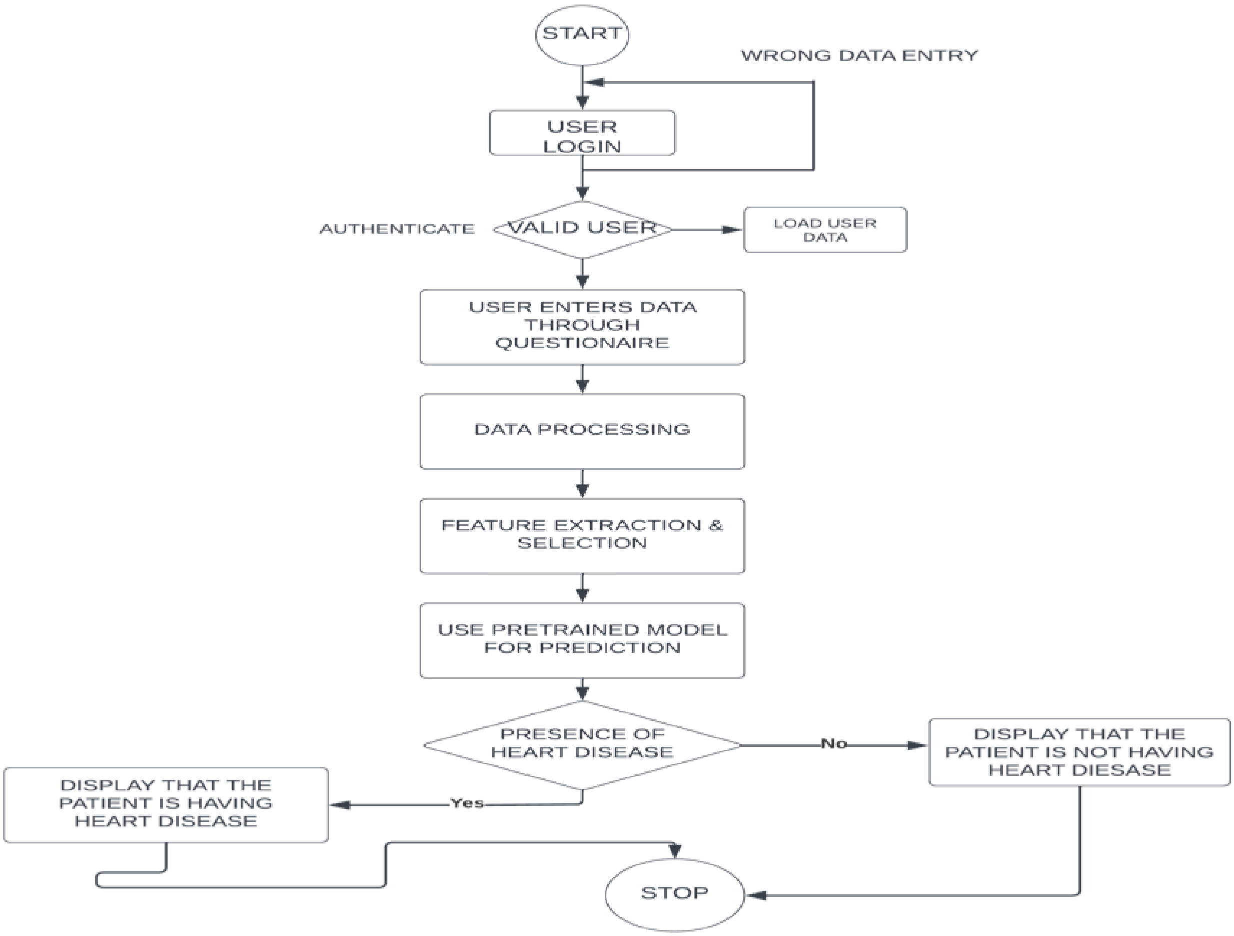

Figure 4.5: Activity Diagram

4.1.5 Sequence Diagram

A sequence diagram is a type of UML diagram that shows the interactions between objects in a system and the sequence of messages exchanged between them. It represents the behavior of a system as a series of interactions between objects, arranged in time order. Sequence diagrams are used to model the interactions between objects in a system and to capture the logic of a business process.

A sequence diagram consists of the following elements:

- Objects: Represented by horizontal arrows, objects are the entities that participate in the interactions.
- Lifelines: Represented by vertical lines, lifelines show the existence of objects over time.

- Messages: Represented by horizontal arrows labeled with the name of the message, messages show the interactions between objects and the sequence in which they occur.

- Activation bars: Represented by horizontal bars on the lifelines, activation bars show the duration of an object's participation in an interaction.

- Return messages: Represented by dotted arrows, return messages show the response of an object to a message.

Sequence diagrams are useful for modeling the interactions between objects in a system and for understanding the logic of a business process. They can be used to identify the roles played by objects in the process, to understand the relationships between objects, and to optimize the process for efficiency and effectiveness.

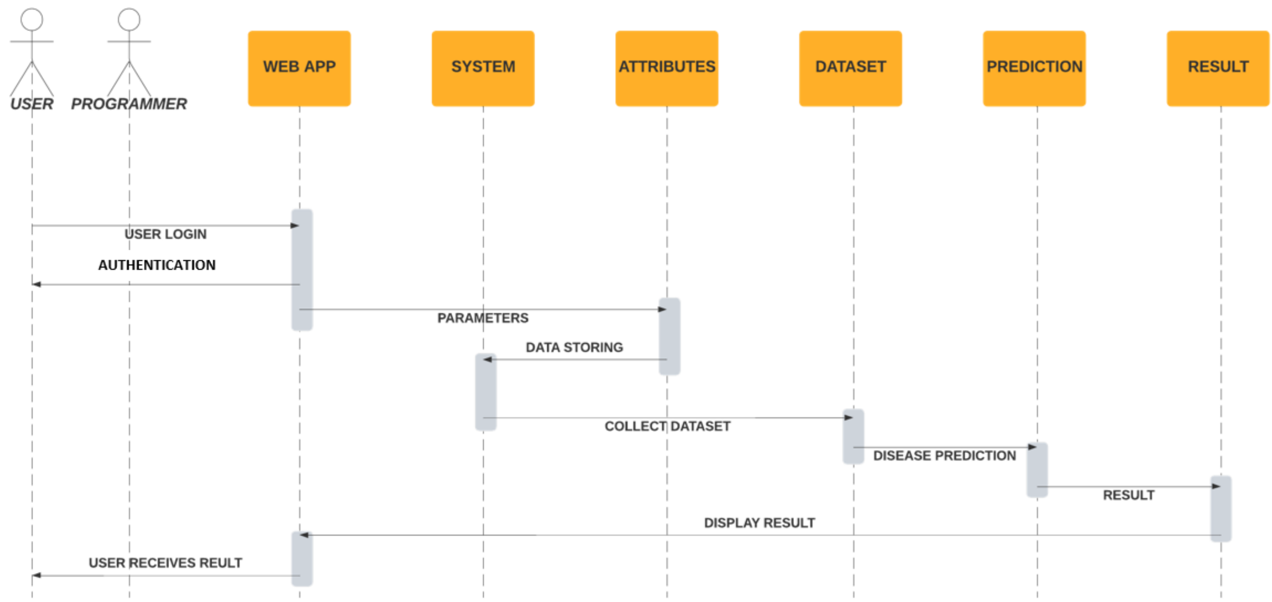

Figure 4.6: Sequence Diagram

4.1.6 Class Diagram

The class diagram is the main building block of object-oriented modeling. It is used for general conceptual modeling of the structure of the application, and for detailed modeling, translating the models into programming code. Class diagrams can also be used for data modeling. UML diagrams like activity diagrams and sequence diagrams can only give the sequence flow of the application, however, the class diagram is a bit different. It is the most popular UML diagram in the coder community.

A class diagram describes the attributes and operations of a class and also the constraints imposed on the system. Class diagrams are widely used in the modeling of object-oriented systems because they are the only UML diagrams, which can be mapped directly with object-oriented languages. The class diagram shows a collection of classes, interfaces, associations, collaborations, and constraints. It is also known as a structural diagram. The purpose of a class diagram is to model the static view of an application. Class diagrams are the only diagrams which can be directly mapped with object-oriented languages and thus widely used at the time of construction.

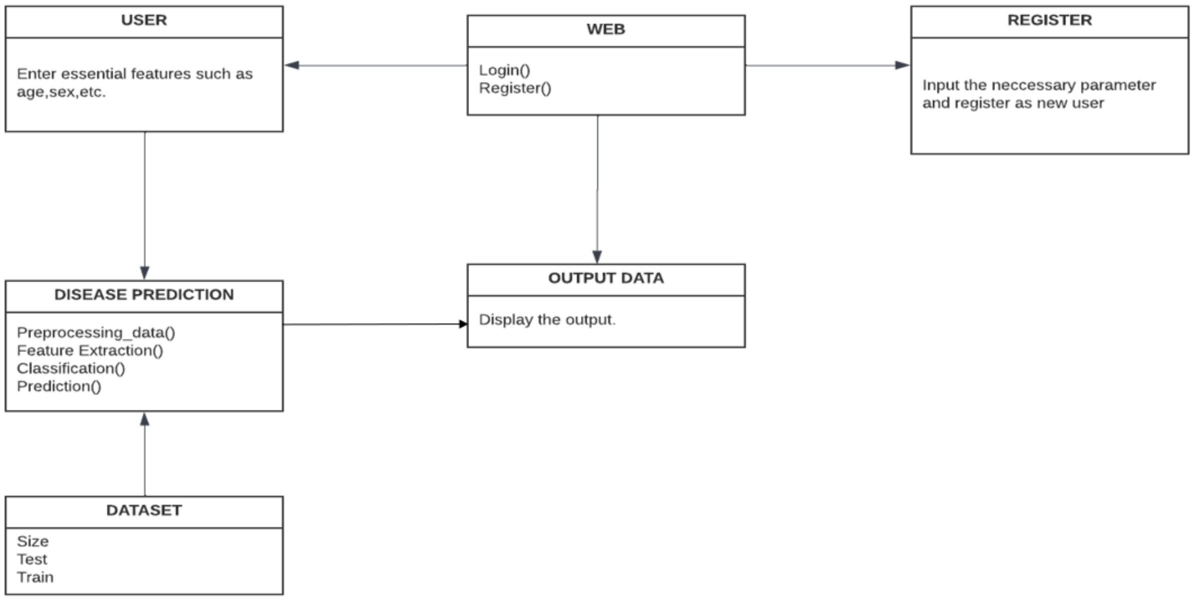

Figure 4.7: Class Diagram

Chapter 5

Preliminary Analysis

5.1 System Requirements

5.1.1 Hardware Requirements

- **Ram:** 4 GB and above
- **Hard disk:** 256 GB and above
- **Processor:** Intel Core i3/Ryzen
- **Processor speed:** 1.60 Hz

5.1.2 Software Requirements

Python

Python is a popular programming language for machine learning. It offers various libraries and frameworks that facilitate the process.Using libraries like Scikit-learn, the Python code can use different machine learning algorithm. The code can further amnalyze and serialize data to improve the accuracy of machine learning model. Python's simplicity and extensive libraries make it a suitable choice for machine learning. Python is used for both model building and we app development using Django.

Google colab

Google Colab is a cloud-based Jupyter Notebook environment provided by Google. It allows users to write, execute, and share Python code in a browser interface. With Google Colab, you can easily access and use powerful hardware resources, including GPUs and TPUs, to accelerate your computations. It provides pre-installed libraries and supports the installation of additional packages. Colab notebooks can be saved in Google Drive and shared with others for collaboration. It also integrates with other Google services, such as Google Sheets and Google BigQuery, making it convenient for data analysis and machine learning tasks.

VS Code

Visual Studio Code (VS Code) is a highly popular and versatile source code editor developed by Microsoft. It offers a rich and intuitive user interface that allows developers to write, edit, and debug code efficiently. One of its key strengths lies in its extensibility, allowing users to customize their coding experience through a vast library of extensions available in the Visual Studio Code Marketplace. These extensions cover a wide range of functionalities, including support for various programming languages, frameworks, and tools. VS Code also includes features like syntax highlighting, code completion, and IntelliSense, which intelligently suggests code completions based on context. It further facilitates collaboration through integrated Git version control and provides seamless integration with build and deployment tools. Whether you are a beginner or an experienced developer, VS Code provides a powerful and flexible environment that can be tailored to meet your specific coding needs.

Django

Django is a popular open-source web framework written in Python. It follows the model-view-controller (MVC) architectural pattern and aims to simplify web development by providing a high-level, robust, and efficient framework. Django offers a set of pre-built components and tools that allow developers to quickly build dynamic web applications.Key features of Django include an object-relational mapper (ORM) that facilitates database interactions, a URL routing system for handling incoming requests, a templating engine for generating dynamic HTML pages, and a powerful administrative interface for managing site content. It also includes built-in support for user authentication, form handling, and security measures, making it a secure choice for web development.

Chapter 6

Implementation

6.1 Collection of data

Initially, we collect a dataset for our heart disease prediction system. After the collection of the dataset, we split the dataset into training data and testing data. The training dataset is used for prediction model learning and testing data is used for evaluating the prediction model. For this project, 75 % of training data is used and 25% of data is used for testing. The dataset used for this project is Heart Disease prediction. The dataset consists of details of 10026 persons out of which, 14 attributes are used for the system.

1	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
2	52	1	0	125	212	0	1	168	0	1	2	2	3	0
3	53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
4	70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
5	61	1	0	148	203	0	1	161	0	0	2	1	3	0
6	62	0	0	138	294	1	1	106	0	1.9	1	3	2	0
7	58	0	0	100	248	0	0	122	0	1	1	0	2	1
8	58	1	0	114	318	0	2	140	0	4.4	0	3	1	0
9	55	1	0	160	289	0	0	145	1	0.8	1	1	3	0
10	46	1	0	120	249	0	0	144	0	0.8	2	0	3	0
11	54	1	0	122	286	0	0	116	1	3.2	1	2	2	0
12	71	0	0	112	149	0	1	125	0	1.6	1	0	2	1
13	43	0	0	132	341	1	0	136	1	3	1	0	3	0
14	34	0	1	118	210	0	1	192	0	0.7	2	0	2	1
15	51	1	0	140	298	0	1	122	1	4.2	1	3	3	0
16	52	1	0	128	204	1	1	156	1	1	1	0	0	0
17	34	0	1	118	210	0	1	192	0	0.7	2	0	2	1
18	51	0	2	140	308	0	0	142	0	1.5	2	1	2	1
19	54	1	0	124	266	0	0	109	1	2.2	1	1	3	0
20	50	0	1	120	244	0	1	162	0	1.1	2	0	2	1
21	58	1	2	140	211	1	0	165	0	0	2	0	2	1
22	60	1	2	140	185	0	0	155	0	3	1	0	2	0
23	67	0	0	106	223	0	1	142	0	0.3	2	2	2	1

Figure 6.1: Dataset

6.2 Selection of attributes

Attribute or Feature selection includes the selection of appropriate attributes for the prediction system. This is used to increase the efficiency of the system. Various attributes of the patient like gender, chest pain type, fasting blood pressure, serum cholesterol, exang, etc are selected for the prediction. The Correlation matrix is used for attribute selection for this model. The backward selection method is used for attribute selection.

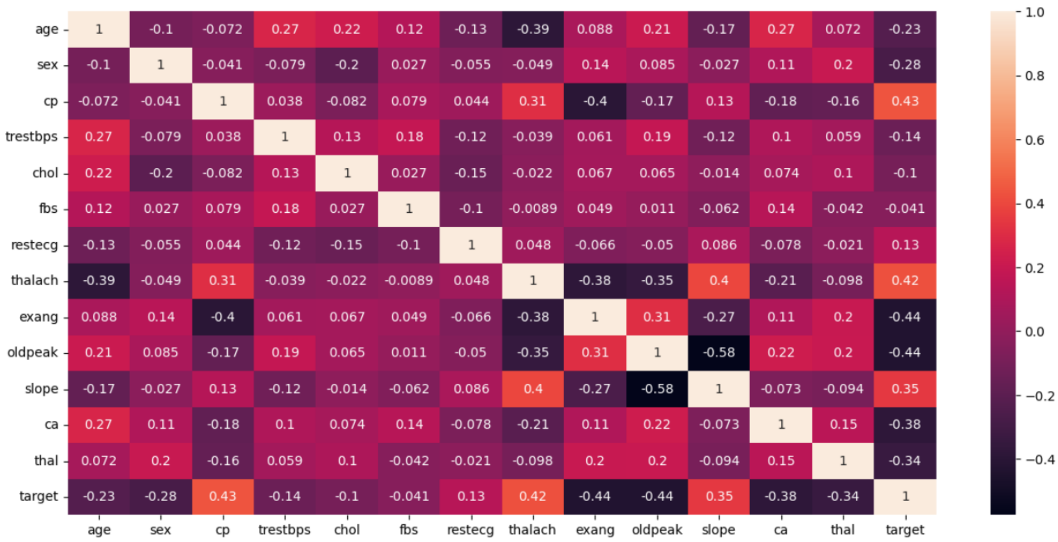

Figure 6.2: correlation matrix

6.3 Preprocessing of data

Data pre-processing is a process of preparing the raw data and making it suitable for a machine learning model. It is the first and crucial step while creating a machine-learning model. This phase of the model handles inconsistent data in order to get more accurate and precise results. This dataset contains missing values. Data pre-processing has the activities like importing datasets, splitting datasets, attribute scaling, Reduction etc. Preprocessing of data is required for improving the accuracy of the model. Then we scale the dataset to normalize all values.

6.4 Data visualization

Data visualization is the representation of data through the use of common graphics, such as charts, plots, infographics, and even animations. These visual displays of information communicate complex data relationships and data-driven insights in a way that is easy to understand. Data visualization is done based on each attribute for data insights. To implement this, we use the python data visualization tool as well as a library function “matplotlib “ to plot a bar graph of discrete variable counts. Analyzing the features of each attribute will be time-consuming. The better option is here to use the Seaborn library and plot all the graphs in a single run using subplots. A summary is formed from the data analysis and visualization for each attribute of data.

6.5 Prediction of heart diseases

The final step is to integrate the heart disease prediction model into the web page. Here users need to input the required data. Various machine learning algorithms like SVM, Naive Bayes, Decision Tree, Random Tree, Logistic Regression, Ada-boost, and Xg-boost are used for classification. Comparative analysis is performed among algorithms and the algorithm that gives the highest accuracy is used for heart disease prediction. Here when the user inputs the data the system will predict the result using the Decision tree algorithm and it gives results such as whether heart disease is affected or not.

6.6 Web Application

Here we created a web application that will predict the patient status automatically by clicking a predict button. Here the prediction model is integrated into the web application.In this web application user need to input the required details about the patient. Prediction based on the trained model, here we used a decision tree.

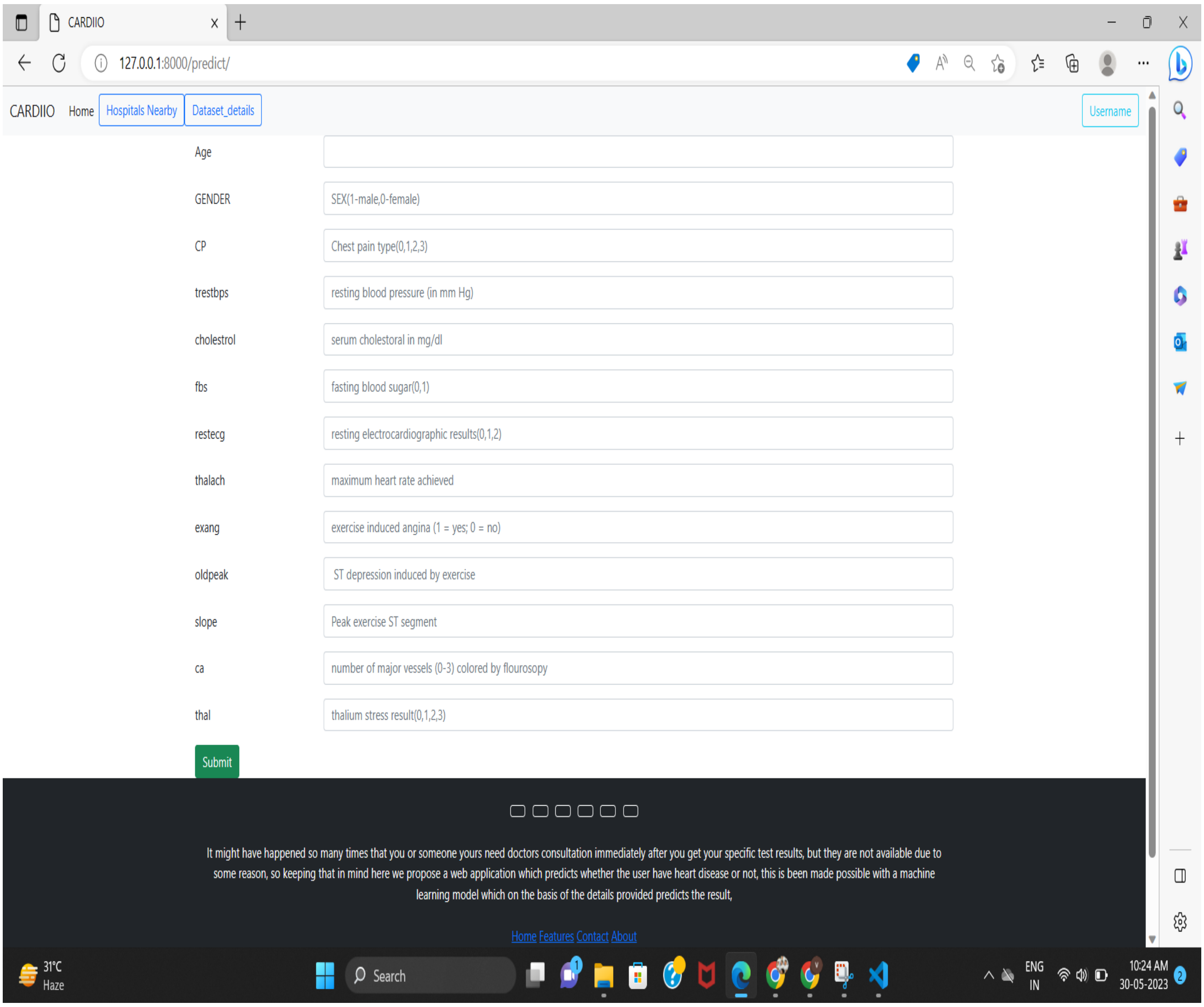

Figure 6.3: Predict page

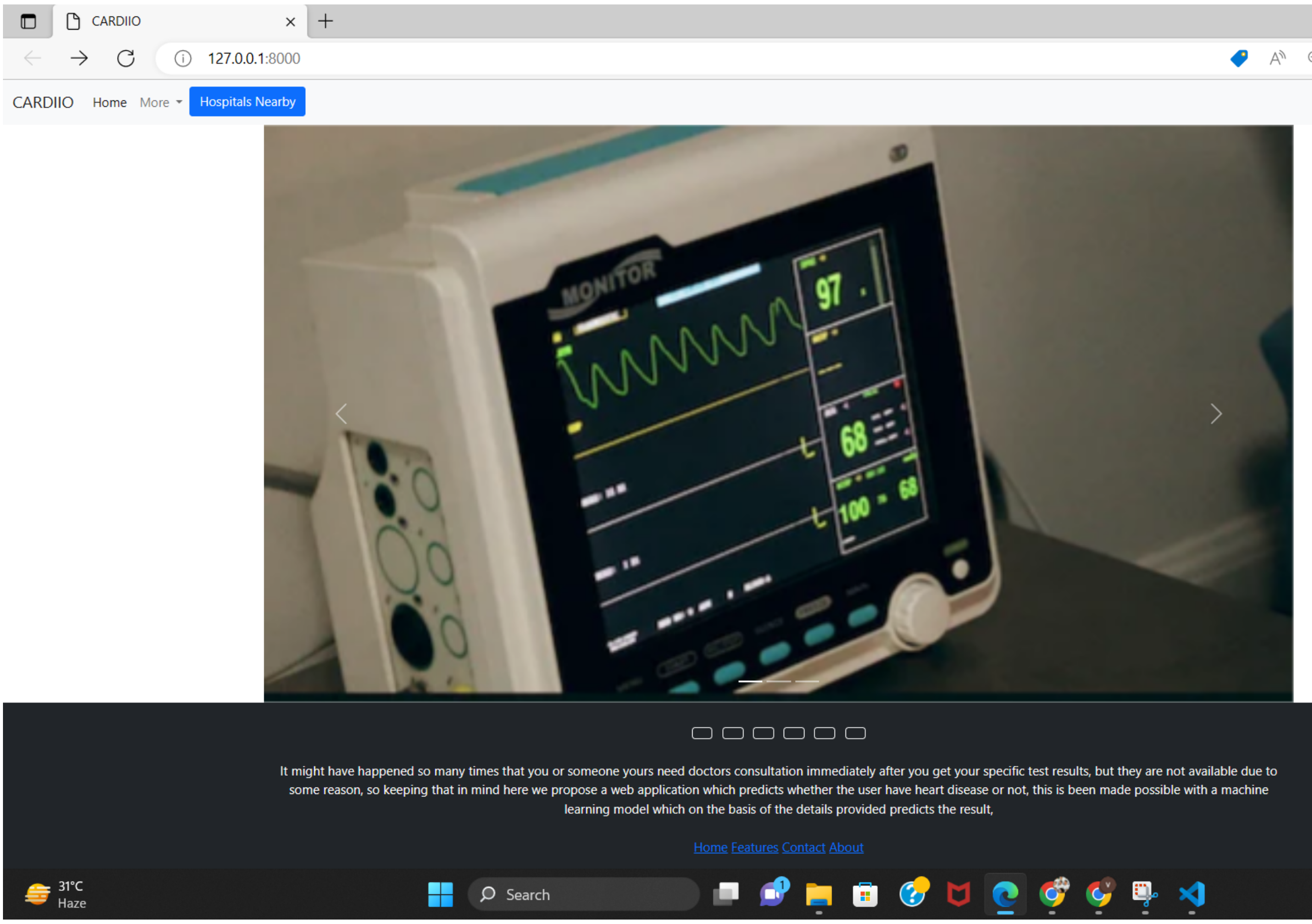

Figure 6.4: homepage

Chapter 7

Result Analysis

After performing the machine learning approach for training and testing we find that accuracy of the Decision Tree is better compared to other algorithms. Accuracy is calculated with the support of the confusion matrix of each algorithm, here the number count of TP, TN, FP, and FN is given, and using the equation of accuracy, the value has been calculated. We used algorithms like Logistic regression of accuracy of 80 percentage, Ada Boost of the accuracy of 82 percentage, KNN of the accuracy of 84 percentage, XG Boost of the accuracy of 93 percentage, Light GBM of the accuracy of 94 percentage, Decision tree of the accuracy of 96.4 percentage. After analysing the accuracy, we have selected the Decision Tree as the best model and used it to predict the disease.

SLNO	ALGORITHM	ACCURACY
1	logistic Regression	80.15%
2	ADA Boost	82.87%
3	KNN	84.43%
4	XG Boost	93.77%
5	Light GBM	94.55%
6	Decision tree	96.4%

Figure 7.1: Algorithms and accuracy

Chapter 8

Conclusion

Heart diseases are a major killer in India and throughout the world, increasing day by day. An application like machine learning is a promising technology for the prediction of heart disease. The early prognosis of heart disease can aid in making decisions on lifestyle changes in high-risk patients and in turn reduce the complications. This system can be the greatest milestone in the field of medicine.The number of people facing heart diseases is on a raise each year. This prompts for its early diagnosis and treatment. The utilization of suitable technology support in this regard can prove to be highly beneficial to the medical fraternity and patients.

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